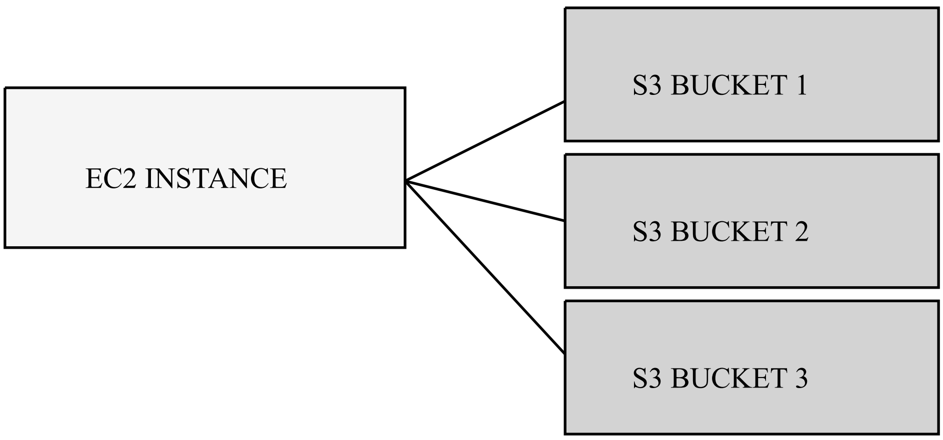
AWS CLOUD PROJECT - EC2 & S3 WEB HOSTING

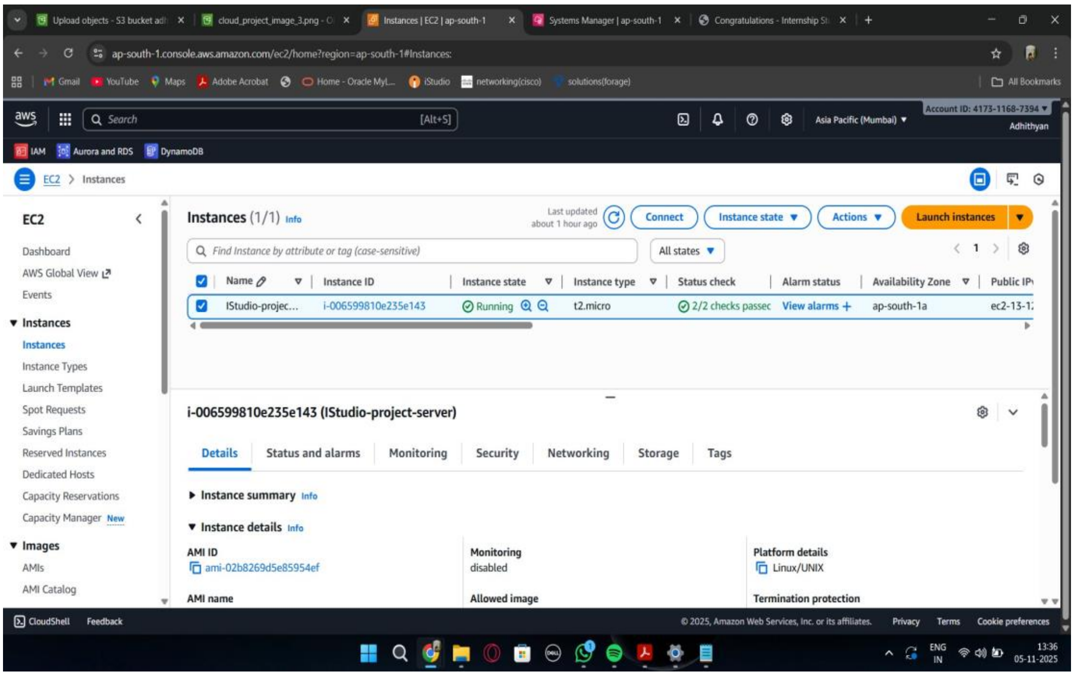
Internship Studio
ADHITHYAN SIVARAMAN T

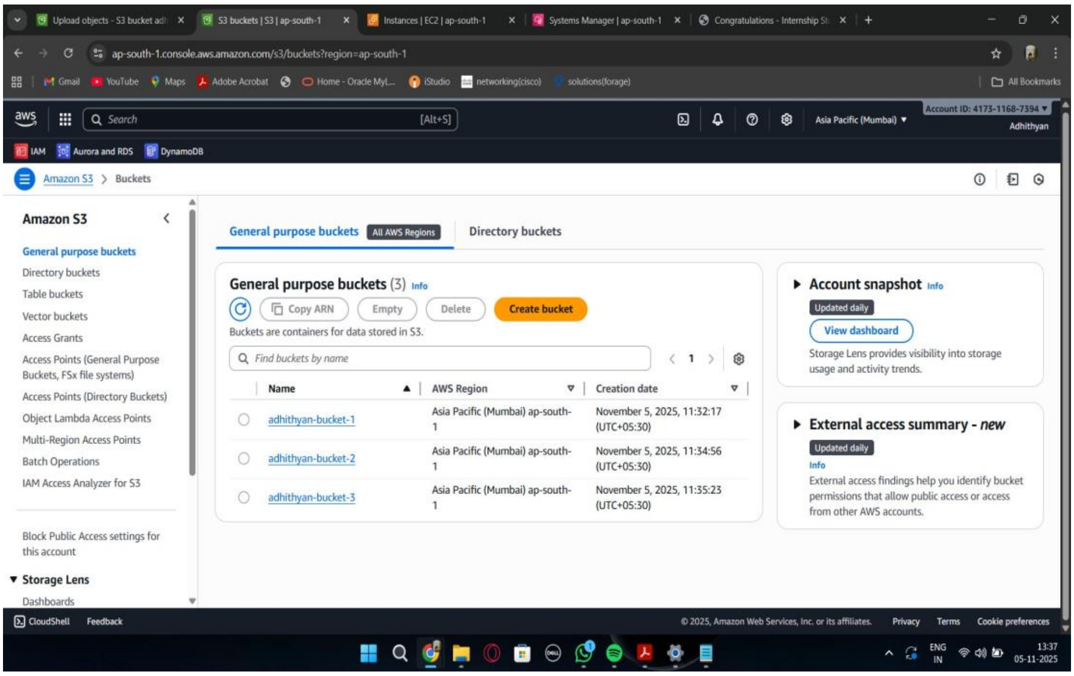
This cloud project demonstrates end-to-end deployment of a web application using Amazon EC2 for compute and Amazon S3 for object storage. The EC2 instance runs a Node.js server, while images are distributed across multiple S3 buckets. AWS Systems Manager (SSM) was used for secure instance access, eliminating the need for key-pair SSH login. Initially, the web application was not reachable through the public IP address. This issue occurred because the Node.js application was listening only on localhost (127.0.0.1). Updating the server to listen on 0.0.0.0 allowed external network access. Additionally, HTTP traffic was enabled through Security Group inbound rules. Another key challenge included configuring S3 buckets for public read access. To resolve this, Block Public Access was disabled and bucket policies were applied to allow GetObject access. This project strengthened understanding of AWS networking, server hosting, IAM permission structures, secure cloud access, and troubleshooting real deployment issues.

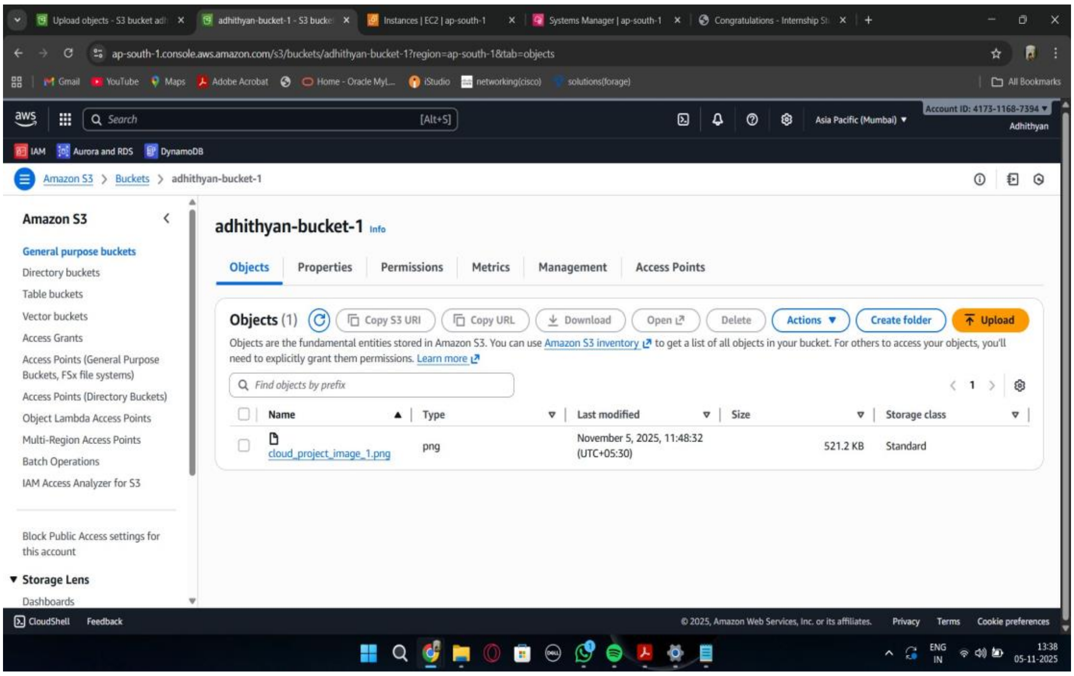
ARCHITECTURE DIAGRAM

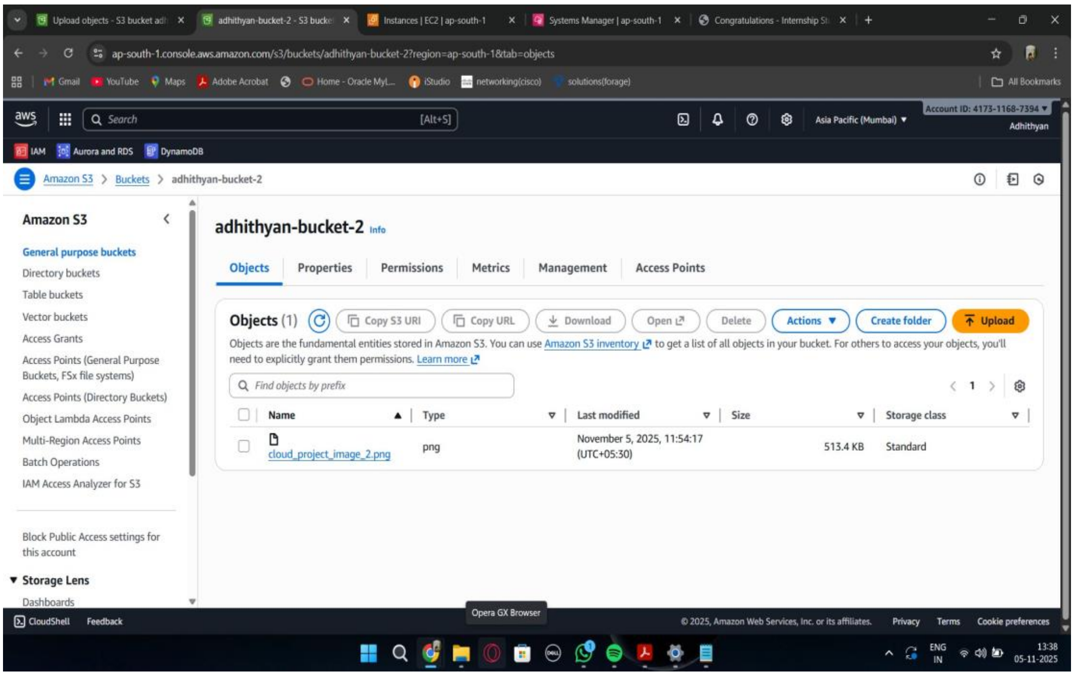


PROJECT EXECUTION SCREENSHOTS






ap-south-1.console.aws.amazon.com/s3/buckets/adhithyan-bucket-1?region=ap-south-1&tab=objects
Amazon S3 > Buckets > adhithyan-bucket-1
Amazon S3
General purpose buckets
Directory buckets
Table buckets
Vector buckets
Access Grants
Access Points (General Purpose Buckets, FSx file systems)
Access Points (Directory Buckets)
Object Lambda Access Points
Multi-Region Access Points
Batch Operations
IAM Access Analyzer for S3
Block Public Access settings for this account
Storage Lens
Dashboards
adhithyan-bucket-1 Info
Objects
Properties
Permissions
Metrics
Management
Access Points
Objects (1)
Copy S3 URI
Copy URL
Download
Open
Delete
Actions
Create folder
Upload
Objects are the fundamental entities stored in Amazon S3. You can use Amazon S3 inventory to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. Learn more
Find objects by prefix
Name
Type
Last modified
Size
Storage class
cloud_project_image_1.png
png
November 5, 2025, 11:48:32 (UTC+05:30)
521.2 KB
Standard
Account ID: 4173-1168-7394
Asia Pacific (Mumbai)
Adhithyan
CloudShell
Feedback
© 2025, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences
13:38
05-11-2025


ap-south-1.console.aws.amazon.com/s3/buckets/adhithyan-bucket-2?region=ap-south-1&tab=objects
Amazon S3 > Buckets > adhithyan-bucket-2
Amazon S3
General purpose buckets
Directory buckets
Table buckets
Vector buckets
Access Grants
Access Points (General Purpose Buckets, FSx file systems)
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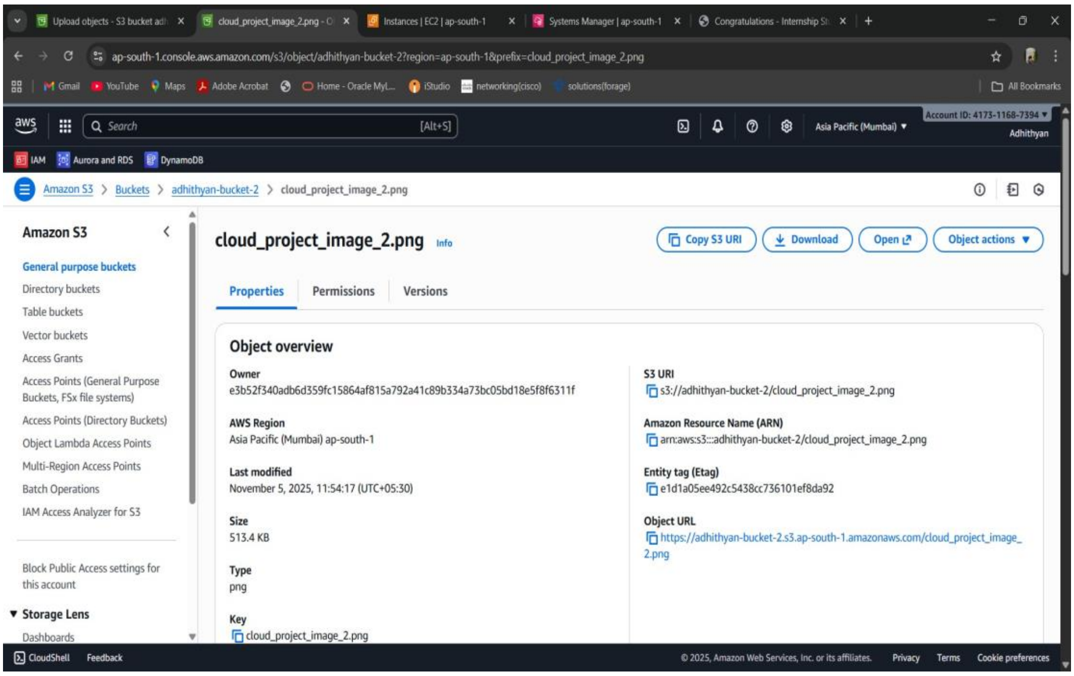
ap-south-1.console.aws.amazon.com/s3/object/adhithyan-bucket-2?region=ap-south-1&prefix=cloud_project_image_2.png
Amazon S3 > Buckets > adhithyan-bucket-2 > cloud_project_image_2.png
Amazon S3
General purpose buckets
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Storage Lens
Dashboards
cloud_project_image_2.png Info
Copy S3 URI
Download
Open
Object actions
Properties
Permissions
Versions
Object overview
Owner
e3b52f340adb6d359fc15864af815a792a41c89b334a73bc05bd18e5f8f6311f
AWS Region
Asia Pacific (Mumbai) ap-south-1
Last modified
November 5, 2025, 11:54:17 (UTC+05:30)
Size
513.4 KB
Type
png
Key
cloud_project_image_2.png
S3 URI
s3://adhithyan-bucket-2/cloud_project_image_2.png
Amazon Resource Name (ARN)
arn:aws:s3:::adhithyan-bucket-2/cloud_project_image_2.png
Entity tag (Etag)
e1d1a05ee492c5438cc736101ef8da92
Object URL
https://adhithyan-bucket-2.s3.ap-south-1.amazonaws.com/cloud_project_image_2.png
CloudShell
Feedback
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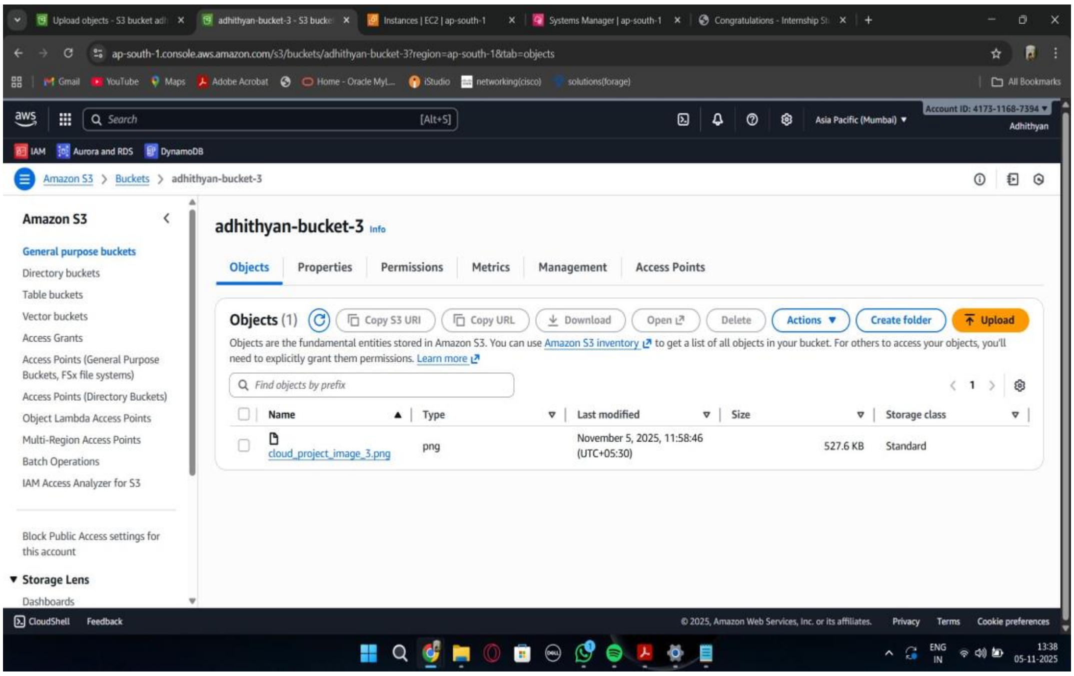
ap-south-1.console.aws.amazon.com/s3/buckets/adhithyan-bucket-3?region=ap-south-1&tab=objects
Amazon S3 > Buckets > adhithyan-bucket-3
adhithyan-bucket-3 Info
Objects
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cloud_project_image_3.png
png
November 5, 2025, 11:58:46 (UTC+05:30)
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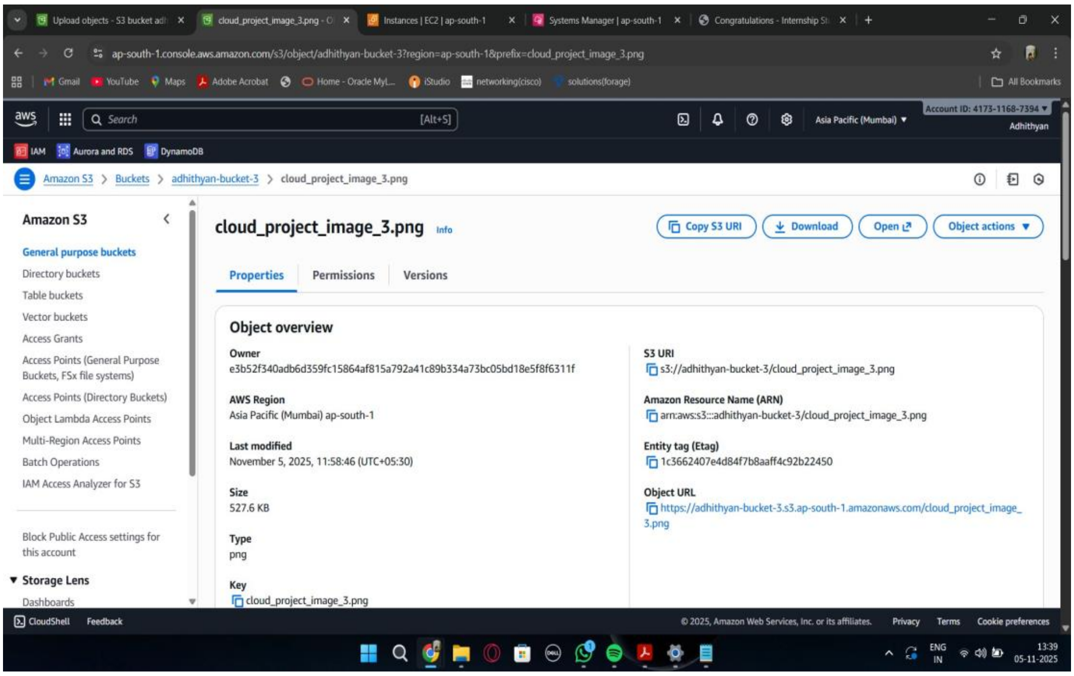
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Owner
e3b52f340adb6d359fc15864af815a792a41c89b334a73bc05bd18e5f8f6311f
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Asia Pacific (Mumbai) ap-south-1
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S3 URI
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Amazon Resource Name (ARN)
arn:aws:s3:::adhithyan-bucket-3/cloud_project_image_3.png
Entity tag (Etag)
1c3662407e4d84f7b8aaff4c92b22450
Object URL
https://adhithyan-bucket-3.s3.ap-south-1.amazonaws.com/cloud_project_image_3.png
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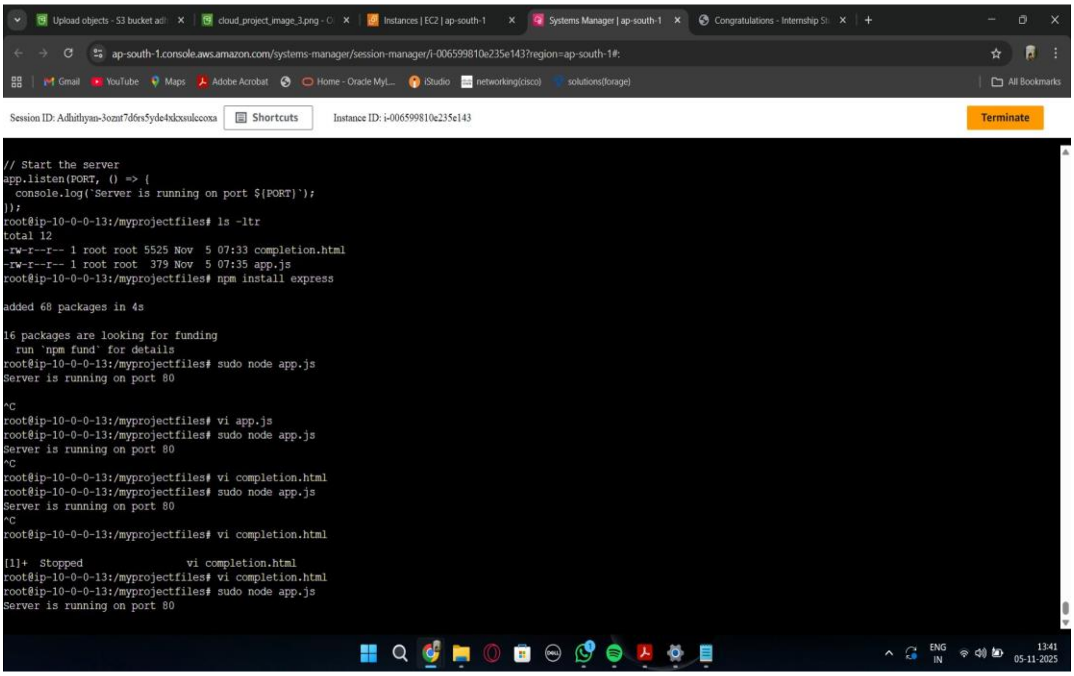
ap-south-1.console.aws.amazon.com/systems-manager/session-manager/i-006599810e235e143?region=ap-south-1#
Instance ID: i-006599810e235e143
Terminate
// Start the server
app.listen(PORT, () => {
console.log(`Server is running on port ${PORT}`);
});
root@ip-10-0-0-13:/myprojectfiles# ls -ltr
total 12
-rw-r--r-- 1 root root 5525 Nov 5 07:33 completion.html
-rw-r--r-- 1 root root 379 Nov 5 07:35 app.js
root@ip-10-0-0-13:/myprojectfiles# npm install express
added 68 packages in 4s
16 packages are looking for funding
run `npm fund` for details
root@ip-10-0-0-13:/myprojectfiles# sudo node app.js
Server is running on port 80
^C
root@ip-10-0-0-13:/myprojectfiles# vi app.js
root@ip-10-0-0-13:/myprojectfiles# sudo node app.js
Server is running on port 80
^C
root@ip-10-0-0-13:/myprojectfiles# vi completion.html
root@ip-10-0-0-13:/myprojectfiles# sudo node app.js
Server is running on port 80
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root@ip-10-0-0-13:/myprojectfiles# vi completion.html
[1]+ Stopped   vi completion.html
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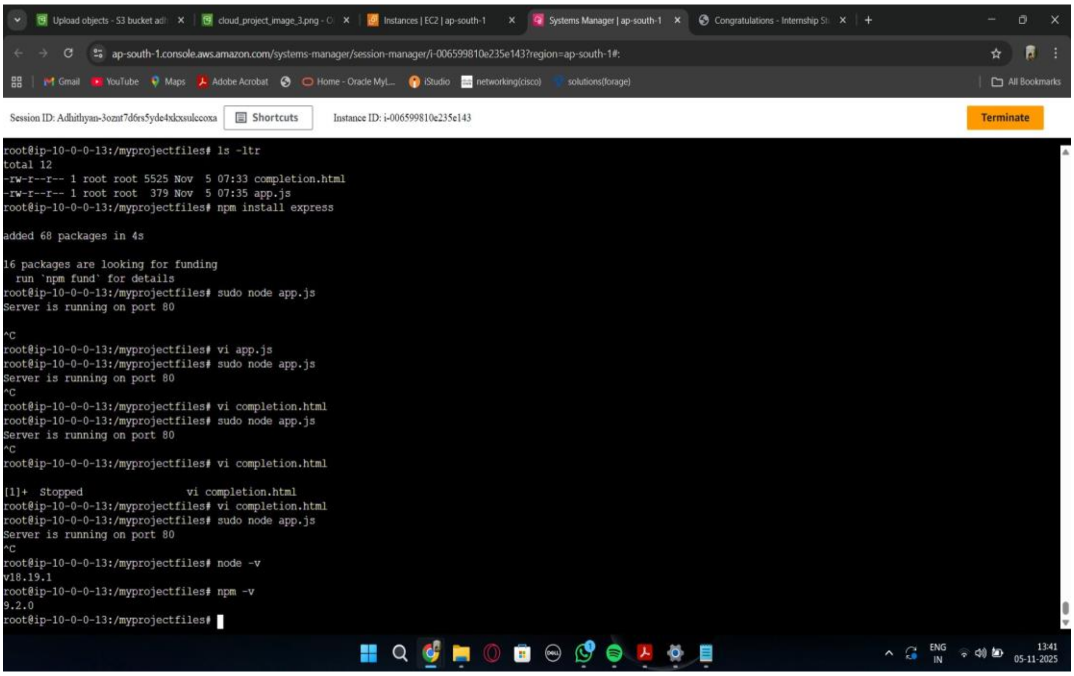
Server is running on port 80


Session ID: Adhithyan-3oznt7d6rs5yde4xlcxsulccoxa
Shortcuts
Instance ID: i-006599810e235e143
Terminate
root@ip-10-0-0-13:/myprojectfiles# ls -ltr
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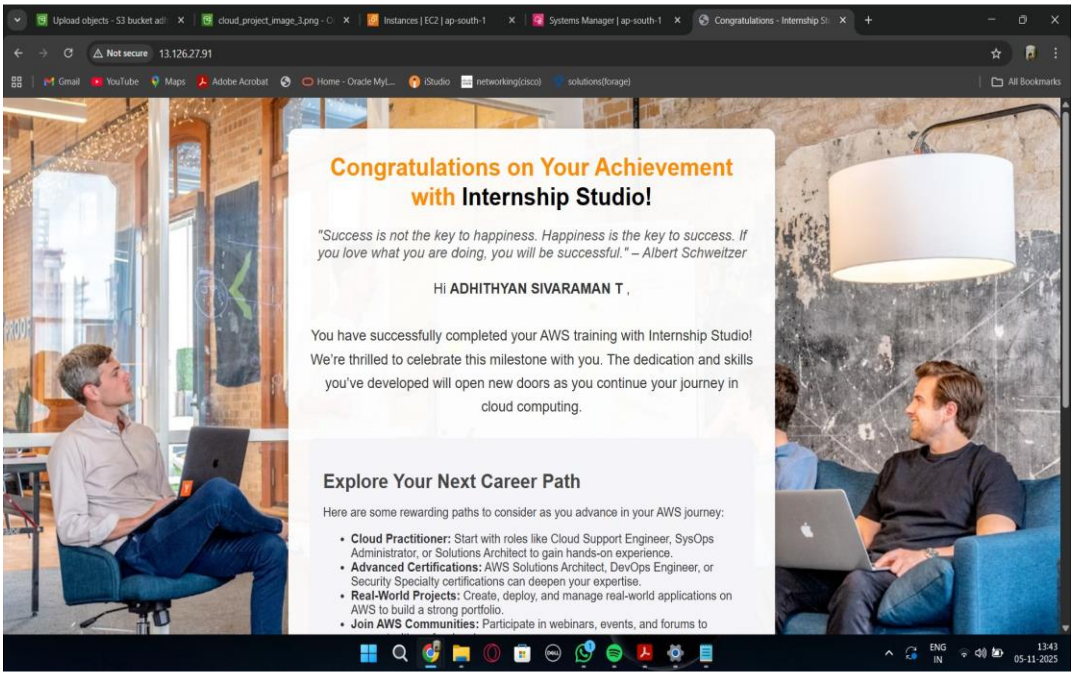
[1]+ Stopped   vi completion.html
root@ip-10-0-0-13:/myprojectfiles# vi completion.html
root@ip-10-0-0-13:/myprojectfiles# sudo node app.js
Server is running on port 80
^C
root@ip-10-0-0-13:/myprojectfiles# node -v
v18.19.1
root@ip-10-0-0-13:/myprojectfiles# npm -v
9.2.0
root@ip-10-0-0-13:/myprojectfiles#
13:41
05-11-2025


Not secure
13.126.27.91
Congratulations on Your Achievement with Internship Studio!
"Success is not the key to happiness. Happiness is the key to success. If you love what you are doing, you will be successful." – Albert Schweitzer
Hi ADHITHYAN SIVARAMAN T ,
You have successfully completed your AWS training with Internship Studio! We're thrilled to celebrate this milestone with you. The dedication and skills you've developed will open new doors as you continue your journey in cloud computing.
Explore Your Next Career Path
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Cloud Practitioner: Start with roles like Cloud Support Engineer, SysOps Administrator, or Solutions Architect to gain hands-on experience.
Advanced Certifications: AWS Solutions Architect, DevOps Engineer, or Security Specialty certifications can deepen your expertise.
Real-World Projects: Create, deploy, and manage real-world applications on AWS to build a strong portfolio.
Join AWS Communities: Participate in webinars, events, and forums to
13:43
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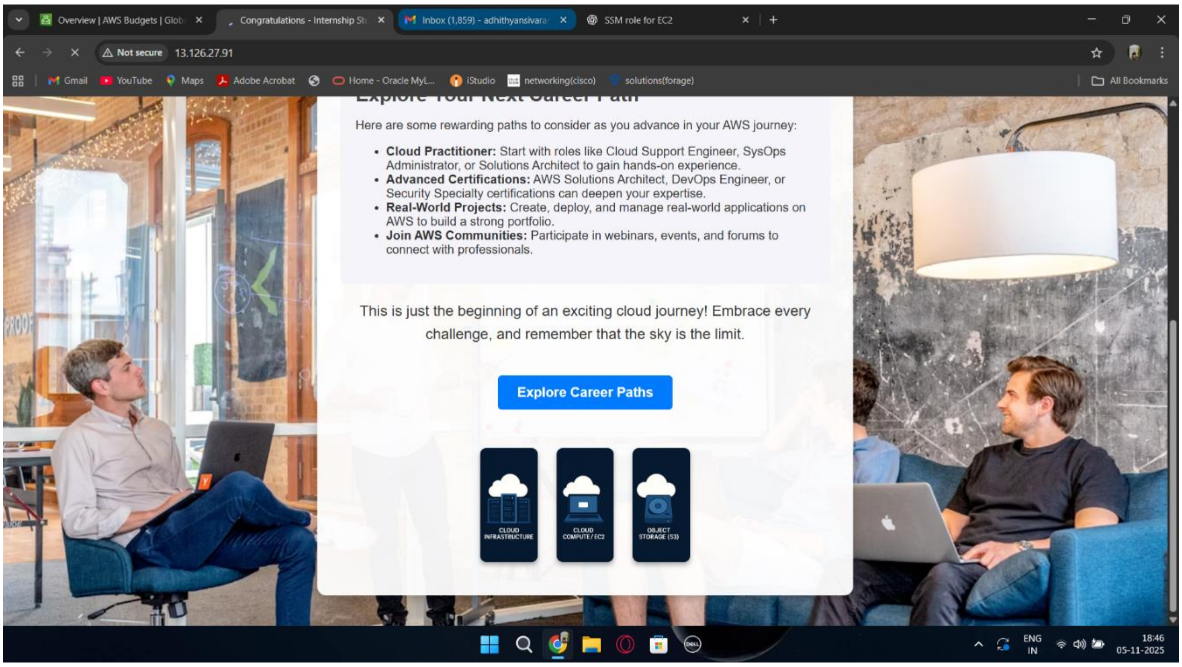

The above screenshot shows the successfully deployed web application hosted on an Amazon EC2 instance. The webpage is publicly accessible via the EC2 public IP address and confirms the correct functioning of the Node.js server along with S3-hosted assets.

CONCLUSION

This project provided hands-on experience in deploying real-world applications using cloud infrastructure. It helped develop a deep understanding of EC2 instance management, secure SSH-less access with SSM Session Manager, Static asset hosting through S3, IAM role assignment, bucket policy configuration, and inbound firewall rule handling. Troubleshooting connectivity issues allowed practical learning of how cloud networking and public accessibility differ from local development environments. Overall, this project builds a strong foundation for advanced cloud architecture, DevOps pipelines, container deployment (ECS/EKS/Docker), automated CI/CD workflows, and scalable production-grade distributed systems.